

Artificial Intelligence & Machine Learning – Beginner Notes

1. Basics of Artificial Intelligence and Machine Learning

What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a technology that enables computers or machines to **perform tasks that normally require human intelligence**.

In simple words:

AI = Making machines behave intelligently.

AI systems can perform tasks such as:

- Understanding language
- Recognizing images
- Making decisions
- Solving problems
- Learning from information
- Recognizing speech

Examples of AI

AI Application	What it does
ChatGPT	Understands and generates text

Google Maps	Finds routes and predicts traffic
Face Unlock	Recognizes your face
Voice Assistants	Understand speech
Recommendation Systems	Suggests movies/products
Self-driving cars	Understand surroundings and make driving decisions

Simple Example

When your phone recognizes your face:

Camera → Captures face → AI analyzes face → Matches stored face → Unlocks phone

What is Machine Learning (ML)?

Machine Learning (ML) is a branch of AI where computers **learn patterns from data and use those patterns to make predictions or decisions.**

In traditional programming:

Data + Rules → Output

In Machine Learning:

Data + Output/Examples → Learn Pattern → Prediction

Example

Suppose we give an ML model information about houses:

Area	Bedrooms	Price
1000 sq.ft	2	₹40 lakh
1500 sq.ft	3	₹60 lakh
2000 sq.ft	4	₹80 lakh

The ML model learns the relationship between **house features and price**.

Then, if we give:

2500 sq.ft + 4 bedrooms

the model can **predict the price**.



Basic Machine Learning Process

Collect Data



Clean Data



Prepare Data



Train ML Model



Test Model



Evaluate Model



Make Predictions

Important Terms

Data: Information used by the machine learning system.

Feature: Input information used for prediction.

Example:

Age

Salary

Tenure

Monthly Charges

These can be features in a customer churn model.

Target: The value we want to predict.

Example:

Churn = Yes / No

Model: The mathematical/statistical system that learns patterns from data.

Training: Teaching the model using existing data.

Testing: Checking how well the trained model works on unseen data.

Prediction: Output generated by the trained model.

2. Difference Between AI, ML, and Deep Learning

This is one of the **most important concepts** to understand.

Relationship

Artificial Intelligence (AI)

└─ Machine Learning (ML)

└─ Deep Learning (DL)

So:

Deep Learning is a part of Machine Learning, and Machine Learning is a part of AI.

AI

AI is the **largest concept**.

It focuses on making machines perform intelligent tasks.

Example:

A robot that can understand commands, navigate, and make decisions.

ML

ML is a **subset of AI**.

Instead of explicitly programming every rule, the machine learns patterns from data.

Example:

Give the model thousands of emails:

Email → Spam
Email → Not Spam
Email → Spam
...

The model learns patterns and predicts whether a new email is spam.

Deep Learning

Deep Learning is a **subset of ML** that uses **artificial neural networks with multiple layers** to learn complex patterns.

It is particularly useful for areas such as:

- Image recognition
- Speech recognition
- Natural language processing
- Computer vision
- Generative AI

Example

For face recognition:

Image
↓

Neural Network
↓
Learn facial patterns
↓
Recognize person



AI vs ML vs DL

Feature	AI	ML	Deep Learning
Meaning	Artificial Intelligence	Machine Learning	Deep Learning
Scope	Broadest	Subset of AI	Subset of ML
Main idea	Intelligent machines	Learn from data	Learn complex patterns
Data requirement	Varies	Usually needs data	Often requires large datasets
Human feature engineering	May be required	Often important	Often reduced
Example	Intelligent robot	Spam detection	Face recognition



Easy Memory Trick

Remember:

AI → Intelligence

ML → Learning from Data

DL → Learning with Deep Neural Networks

3. Real-World Applications of AI

AI is already used in many industries.



1. Healthcare

AI can help with:

- Medical image analysis
- Disease-risk prediction
- Patient monitoring
- Drug discovery

Example

AI can analyze medical images and help identify patterns that may require further examination.



2. Banking and Finance

AI/ML is used for:

- Fraud detection
- Credit scoring
- Risk analysis
- Customer churn prediction
- Transaction monitoring

Example

If a customer's card suddenly makes unusual transactions, an ML system can identify the transaction as potentially fraudulent.



3. E-Commerce

AI is used for:

- Product recommendations
- Search
- Customer behavior analysis
- Demand forecasting
- Chatbots

Example

When an online shopping website recommends:

"You may also like..."

that recommendation can be generated using recommendation algorithms.

4. Transportation

AI is used for:

- Route optimization
- Traffic prediction
- Driver assistance
- Autonomous driving research

Example

Google Maps predicts traffic and suggests alternative routes.

5. Entertainment

AI is used by platforms to recommend:

- Movies
- Music
- Videos
- Shows

Example:

Your previous activity



AI/ML system



Learn your preferences



Recommend content

6. Chatbots and Generative AI

AI can be used to:

- Answer questions
- Generate text
- Summarize information
- Translate languages
- Generate code
- Create images

Examples include AI assistants and conversational systems.

7. Cybersecurity

AI/ML can help detect:

- Suspicious activity
 - Malware
 - Unusual network behavior
 - Potential attacks
-

4. Install Python and Practice Programming Concepts

What is Python?

Python is a high-level, general-purpose programming language.

It is popular in:

- Artificial Intelligence
- Machine Learning
- Data Science
- Web Development
- Automation
- Software Development

One reason beginners like Python is its relatively simple syntax.

Installing Python

You can install Python from the official Python website:

[Python Official Website](https://www.python.org/)

After installation, open **Command Prompt** and check:

```
python --version
```

or:

```
py --version
```

You should see a Python version number.

Python Basics

4.1 Variables

A variable stores a value.

```
name = "Lokesh"  
age = 21  
marks = 85.5
```

Here:

```
name → "Lokesh"  
age → 21  
marks → 85.5
```

Printing variables

```
name = "Lokesh"  
print(name)
```

Output:

Lokesh

4.2 Data Types

Common Python data types:

Data Type	Example
-----------	---------

<code>int</code>	<code>10</code>
------------------	-----------------

<code>float</code>	<code>10.5</code>
--------------------	-------------------

<code>str</code>	<code>"Hello"</code>
------------------	----------------------

<code>bool</code>	<code>True /</code> <code>False</code>
-------------------	---

<code>list</code>	<code>[10, 20,</code> <code>30]</code>
-------------------	---

Example:

```
age = 21
price = 99.5
name = "Lokesh"
student = True
```

4.3 Taking Input

Python can take input from the user using `input()`.

```
name = input("Enter your name: ")
print("Hello", name)
```

Example:

```
Enter your name: Lokesh
Hello Lokesh
```

Important

`input()` normally returns a **string**.

If you want an integer:

```
age = int(input("Enter your age: "))
```

4.4 Operators

Arithmetic Operators

```
a = 10
```

```
b = 3
```

```
print(a + b)
```

```
print(a - b)
```

```
print(a * b)
```

```
print(a / b)
```

```
print(a % b)
```

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Remainder
**	Power

4.5 Conditional Statements

We use `if`, `elif`, and `else` to make decisions.

```
age = 20
```

```
if age >= 18:
```

```
    print("Eligible")
```

```
else:
```

```
    print("Not eligible")
```

Flow

Age

↓
Is age >= 18?
↓
Yes → Eligible
No → Not eligible

4.6 Loops

Loops are used to **repeat a task**.

For Loop

```
for i in range(5):  
    print(i)
```

Output:

```
0  
1  
2  
3  
4
```

Example

```
for i in range(1, 6):  
    print(i)
```

Output:

```
1  
2  
3  
4  
5
```

While Loop

A **while** loop continues while a condition is true.

```
i = 1
```

```
while i <= 5:  
    print(i)  
    i = i + 1
```

Output:

```
1  
2  
3  
4  
5
```

4.7 Lists

A list stores multiple values.

```
numbers = [10, 20, 30, 40]
```

Access an element:

```
print(numbers[0])
```

Output:

```
10
```

Remember:

Python indexing starts from **0**.

Index:	0	1	2	3
Value:	10	20	30	40

Add an item

```
numbers.append(50)
```

Now:

```
[10, 20, 30, 40, 50]
```

Loop through a list

```
for number in numbers:
```

```
print(number)
```

4.8 Functions

A function is a **reusable block of code** that performs a specific task.

```
def greet():  
    print("Hello!")
```

Call the function:

```
greet()
```

Output:

```
Hello!
```

Function with parameters

```
def greet(name):  
    print("Hello", name)
```

```
greet("Lokesh")
```

Output:

```
Hello Lokesh
```

Function with return

```
def add(a, b):  
    return a + b
```

```
result = add(10, 20)
```

```
print(result)
```

Output:

```
30
```

Easy understanding

Input



Function



Processing



Output

5. Beginner-Friendly Python Programs

Now let's practice some simple programs. 🖥️

Program 1: Print Hello World

```
print("Hello World")
```

Output:

Hello World

Program 2: Add Two Numbers

```
a = 10
```

```
b = 20
```

```
sum = a + b
```

```
print("Sum:", sum)
```

Output:

Sum: 30

Program 3: Check Even or Odd

```
number = int(input("Enter a number: "))
```

```
if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Logic

Number
↓
number % 2
↓
0 → Even
1 → Odd

Program 4: Find the Largest of Two Numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
```

```
if a > b:
    print("Largest:", a)
else:
    print("Largest:", b)
```

Program 5: Print Numbers 1 to 10

```
for i in range(1, 11):
    print(i)
```

Program 6: Sum of Numbers from 1 to 10

```
total = 0

for i in range(1, 11):
    total = total + i

print("Total:", total)
```

Output:

Total: 55

How it works

total = 0

i = 1 → total = 1

i = 2 → total = 3

i = 3 → total = 6

...

i = 10 → total = 55

Program 7: Find the Largest Number in a List

```
numbers = [10, 25, 5, 40, 15]
```

```
largest = numbers[0]
```

```
for number in numbers:
```

```
    if number > largest:
```

```
        largest = number
```

```
print("Largest:", largest)
```

Output:

Largest: 40

Program 8: Count Even Numbers

```
numbers = [1, 2, 3, 4, 5, 6]
```

```
count = 0
```

```
for number in numbers:
```

```
    if number % 2 == 0:
```

```
        count = count + 1
```

```
print("Even numbers:", count)
```

Output:

Even numbers: 3



Quick Revision Sheet

Before moving into ML, make sure you understand these:

Topic	What you should understand
AI	Machines performing intelligent tasks
ML	Machines learning patterns from data
Deep Learning	ML using deep neural networks
Variable	Stores a value
Data Type	Type of stored value
<code>if</code>	Makes a decision
<code>for</code>	Repeats code
<code>while</code>	Repeats while condition is true
List	Stores multiple values
Function	Reusable block of code
<code>input()</code>	Takes user input
<code>print()</code>	Displays output



The Big Picture

You can remember the entire learning path like this:

Artificial Intelligence



Machine Learning



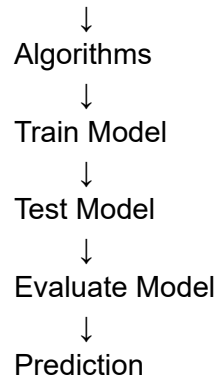
Deep Learning



Python



Data



★ **Most important beginner idea**

Don't try to memorize Python programs line by line.

For every program, ask:

- 1. What is the input?**
- 2. What processing do I need?**
- 3. What condition/loop is required?**
- 4. What should be the output?**

That way, you gradually move from **"I can understand code"** → **"I can write code myself."**